OMAR KHATAB

Frontend Developer — React / Next.js / TypeScript
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Portfolio | GitHub | LinkedIn

SUMMARY

Frontend Developer with a Mechanical Power Engineering background. Built and deployed projects with React, Next.js, and TypeScript, including an HVAC calculator applying engineering formulas and custom Shopify sections.

TECHNICAL SKILLS

Languages & Core: JavaScript (ES6+), TypeScript, HTML5, CSS3
Frameworks & Libraries: React.js, Next.js (App Router, Server Components), Material UI (MUI)
E-Commerce / CMS: Shopify (Liquid, Ajax Cart API, Section Rendering API), Shopify CLI
Styling & Animation: Tailwind CSS, Framer Motion, BEM, Flexbox, CSS Grid, Responsive Design (320px–1920px)
State & Data: Context API, REST API integration
Tools & Workflow: Git, GitHub, Vercel, Windsurf (AI-assisted development), PSD to HTML conversion, Pixel-Perfect implementation

PROJECTS (PRACTICAL EXPERIENCE)

HVAC Duct & BTU Calculator (Featured) | Live Demo | Code | Sep 2026
- Built a multi-page engineering tool (Next.js App Router, TypeScript) that estimates BTU cooling load and sizes rectangular ducts, with a transparent "show your work" formula breakdown.
- Structured the calculations as pure functions and built a reusable typed input component (slider + validation) shared across both calculators.

Shopify Theme Development — Custom E-Commerce Sections | Code & Demo | Sep 2026
- Implemented a Figma design as custom sections (Banner + Product Grid with Quick View) on the Dawn theme using Liquid, vanilla JavaScript, and Shopify's Ajax Cart and Section Rendering APIs.
- Built dynamic variant selectors with O(1) hash-map lookup (instead of a linear search), a Black + Medium cross-sell that auto-adds a related product, and a header cart icon that refreshes without a page reload.
- Prevented race conditions in async add-to-cart by locking the popup while a request is in flight; worked through a feature-branch and Pull Request workflow.

E-Commerce Store — React | Live Demo | Code | Jul – Aug 2026
- Built a full-featured storefront with React, React Router, and Context API — including live product data via a REST API (DummyJSON), error handling, and skeleton loading states.
- Persisted cart/favorites with localStorage, added search and category filtering, and used Framer Motion for a mobile-first responsive UI.

Earlier Work — Responsive Pages (HTML5, CSS3, Vanilla JS) | Live Demo | Code | 2021 – 2024
- Built 5 responsive pages and an admin dashboard from PSD designs alongside my studies, adding new skills with each one: semantic HTML5, CSS3 (BEM, Flexbox, Grid) and vanilla JavaScript, no frameworks, before moving to React and Next.js.

ENGINEERING BACKGROUND — MECHANICAL POWER

- **Internships (Aug–Sep 2022):** National Authority for Tunnels (metro HVAC, ventilation, and fire-fighting systems) and Arab Contractors (engine assembly with IC engines).
- **Temperature Control System (2023–2024):** Arduino-based tuned PID controller with a variable-speed fan holding a constant temperature inside a heated enclosure.
- **Solar Cell Cooling System (Graduation Project, 2023–2024):** simulated several cooling designs to pick the best one; measured with temperature sensors, the cooling lowered panel temperature by 10–20°C from about 80°C.
- **Pitch Control System (2021–2022):** one of three subsystems of a wind turbine project; modeled and simulated a blade-pitch controller that keeps rotor speed constant.

LEARNING & DEVELOPMENT

Currently Learning: Advanced React.js (Comprehensive Course, Académie Tarmiz) — In Progress

EDUCATION

BSc, Mechanical Power Engineering — Ain Shams University
Cairo, Egypt | 2019 – 2024 | Military Service: Completed
Languages: Arabic (Native), English (Professional Working Proficiency — technical reading, writing, and documentation)